

Evaluation Report

INFO 90009

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1 Introduction

1.1 Product Under Evaluation

Roota is an interactive prototype built on Figma designed to help newly immigrated families discover, understand and participate in local community activities. Its primary target audience is currently immigrant families with children. It addresses the potential for parents to experience a lack of information, uncertainty and concerns about participation when faced with activities in unfamiliar communities.

The prototype includes several core functions: a homepage for browse community activities, an activity details page, a scenario simulation function, a Friends Participate function and a Kids Mode for children. Different functions support users in assess information, establish psychological expectations. For example, the activity details page displays basic activity information; the scenario simulation function helps users understand possible scenarios during the activity; the Friends Participate function enhances users' sense of social security by showing whether acquaintances or other families are participating; and Kids Mode uses comics or picture book-style content to help children understand the activity process.

According to the Evaluation Plan, Roota's target users are newly immigrated families with children. This evaluation is more like a formative and proof-of-concept.

Therefore, there will be no summative evaluation of the final product.

1.2 Objectives

The purpose of this usability assessment is to understand whether Roota can effectively help new immigrant families understand, navigate and prepare to participate in local community activities. Since Roota is currently in the Figma prototype stage, so this assessment is more focuses on identify usability issues through real user feedback and provide a basis for subsequent design iterations. This assessment focuses on whether users can gain sufficient information and confidence from the prototype to consider participate in community activities. For newly immigrated families, participate in community activities may present challenges such as language barriers, unfamiliarity with the activity process, lack of familiar companions and uncertainty about the suitability of their children. Therefore, this study examines whether Roota can reduce these uncertainties and provide effective assistance in terms of information clarity, social support and children's participation.

1.3 Research Questions

This usability assessment revolves around the following five research questions:

RQ1: Can new immigrant families quickly understand and browse community activity through Roota?

RQ2: Does the activity details page provide enough information for people?

RQ3: Can the scenario simulation reduce users' uncertainty and help users build pre-participation expectations?

RQ4: Does the Friends Participate feature increase users' sense of social security and willingness when participate?

RQ5: Is Kids Mode appeal to children?

These research questions cover activity comprehension, scenario simulation, peer

engagement prompts, child-centered models and parental decision support.

2 Methodology

2.1 Evaluation Type

This study employed a formative usability evaluation. Because Roota is currently in the prototype stage, so this evaluation focuses on whether users can understand the prototype's main functionalities and what usability issues exist in the current design that require improvement.

This evaluation is primarily intended to support subsequent design iterations.

Therefore, the findings in this report do not represent final conclusions.

2.2 Participants

This availability assessment is aimed at adults in newly immigrated families with children. According to the Evaluation Plan, participants need to have a potential need for community participation, be able to use a mobile phone or tablet and willing to sign an informed consent form to participate before the study.

In practice, the team invited 6 adult participants. Because children are minors, so this evaluation did not directly recruit children to participate in the test. Adult participants evaluated the applicability, appeal and potential problems of Kids Mode from the perspective of parents or family members. Therefore, the results of Kids Mode reflect adults' judgments on children's user experience, rather than direct test feedback from child users.

All participants were required to complete the main tasks of the Roota prototype (see below). During the test, participants expressed their immediate thoughts through Think Aloud and completed a brief semi-structured interview after the task.

2.3 Test Method

This assessment primarily employed qualitative research. Core test methods include Think Aloud usability test, semi-structured interviews and behavioral observation.

2.3.1 Think Aloud Usability Test

Think Aloud is the primary methodology for this test. During the test, participants use the Roota's Figma prototype and express their thoughts, including what they were looking at, what are they looking for, which parts is easier understanding, and what parts is confused at the same time.

2.3.2 Semi-structured interview

After completed the task, the researchers conducted a semi-structured interviews with the participants. The interview questions focused on the overall user experience, the most memorable features, the scenario simulation features, the Friends Participate feature, Kids Mode and their future usage intentions.

2.3.3 Behavioral observation

In addition to language feedback, researchers also observed participants' operational behaviors and nonverbal responses during the test. Include pauses, hesitations, repeated viewings, active clicking, page switching methods, ignore buttons or visual elements and changes in interest in certain functions.

2.4 Test Materials

The main materials used in this test included the Figma interactive prototype, test scripts, record equipment, observation sheets, semi-structured interview questions,

Plain Language Statement and Consent Form.

First, the core material is Roota's Figma interactive prototype. This prototype contains the main pages and functions required for this evaluation. It includes the homepage, activity details page, scene simulation function, Friends Participate and Kids Mode.

Secondly, the researchers used a standardized test script to guide participants through the test. The script included an opening statement, a "Think Aloud" prompt, instructions for 5 tasks and a brief post-test interview. Standardized script ensured that different participants received relatively consistent task instructions, then improve the consistency of the testing process.

Third, researchers used recording equipment during the testing process. The recordings were primarily used for subsequent transcription and thematic analysis. Which help researchers to review participants' immediate thoughts, concerns and suggestions during the process. Record was only conducted with the participants' consent.

Fourth, researchers used observation forms to record participants' task completion status, pauses or hesitations, key information, etc. The forms were mainly used to record the completion status of each task, observation content and satisfaction rate. Finally, this test also used Plain Language Statement and Consent Form as ethical materials.

Materials	Use
Figma Interactive Prototype	For participants to complete five usability testing tasks
Test Script	Ensure consistency between test instructions and task guidance
Recording Equipment	Recording Think Aloud and interview content
Observation Form	Record task completion, behavioral observations and key feedback
Semi-structured Interview Questions	Guiding in-depth interviews after the test
Plain Language Statement	Explain the research purpose and rights to participants
Consent Form	Obtain informed consent from participants

2.5 Test Procedure

This test was an individual user test. Each participant completed one full Roota prototype test, which included an opening statement, informed consent confirmation, Think Aloud task testing, a semi-structured interview and a closing statement.

A single test was expected to last approximately 20–25 minutes, including a 3–5 minute introduction, a 10–15 minute task test and a 5–10 minute interview.

2.6 Task

Task 1: Browse community activities

Task Instructions: Please browse the homepage and find a community activity that interests you.

Test Focus:

- Learnability: Can the user understand the browse method without guidance?
- Efficiency: Does the system support quick location?
- Satisfaction: Is the activity display attractive?

Task 2: View Activity Details

Task Instructions: Please click on the activity you just selected and describe what the activity is about, who it is suitable for and what will happen if you attend.

Test Focus:

- Effectiveness: Whether key information about the activity can be fully obtained.
- Errors: Whether there are any misunderstandings or misinterpretations of information.
- Utility: Whether the information meets the needs of participants in the decision-making process.

Task 3: Experience scenario simulation

Task Instructions: Please try out the activity scenario simulation feature and describe your feelings and expectations for the activity.

Test Focus:

- Satisfaction (emotional response): Does it reduce feelings of unfamiliarity and build psychological expectations?
- Learnability: Is the function intuitive and easy to understand?
- Effectiveness: Does it make it clearer what will happen in the activity?

Task 4: Check the Friends Participation prompts

Task Instructions: Please check the "Friends Participate" prompt to see if it will affect your participation decision.

Test Focus:

- Satisfaction (Social Trust): Does it improve a sense of security to you?
- Utility: Does the social function have decision-making value?
- Errors: Is the functional intent easily understood?

Task 5: Experience Kids Mode

Task Instructions: Please try the children's mode and describe which parts you find most like and whether you would be happy to let your child use it.

Test Focus:

- Learnability: Is the interface intuitive and user-friendly?
- Satisfaction: Does the visual design align with children's preferences?
- Effectiveness: Can children independently understand the activity content?

2.7 Data Collection and Analysis

This assessment primarily collected qualitative data, supplemented by simple task completion records and problem frequency statistics.

2.7.1 Data Collection

The data for this test mainly came from 3 sources: Think Aloud, semi-structured interview records and behavioral observations of the researchers during the test.

2.7.2 Data Analysis

This data analysis uses thematic analysis. Researchers first compiled participants' test records and interview transcripts, then repeatedly read the texts, marking segments related to usability issues, positive feedback, user needs and design suggestions. Next, similar content was categorized to form several core themes.

2.7.3 Quantitative Data

This assessment also recorded some basic quantitative information include task completion and the frequency of question occurrence. For example, researchers recorded whether participants completed the 5 main tasks and whether certain questions were repeatedly mentioned by multiple participants.

However, since the primary objective of this assessment was formative design feedback rather than statistical validation, the quantitative data served only as supplementary information.

2.8 Ethical Considerations

This test with basic research ethics requirements. Before the test, all participants were informed of the research objectives, testing procedures, data usage and withdrawal rights. The researchers also stated that this test is for Roota's design, not the participants' abilities. All feedback was neither right nor wrong.

Participants need read and signed the Consent Form before the test. And understood the research content information through the Plain Language Statement. They can stop at any time.

To protect the privacy, this report will not use participants' real names, only use P1-P6 to identify. Recordings and test logs will only be used for course project analysis. In the end, because children are minors, so no children were directly recruited for this assessment. The feedback from Kids Mode came from the perspectives of the parents or family members of adult participants.

2.9 Limitations

This assessment has certain limitations. First, the sample size is relatively small. Only 6 adult participants. This size is still insufficient to represent the usage of the product by all new immigrant families.

Secondly, participants were primarily recruited through snowball sampling. And all participants came from Chinese immigrant families. It may not fully represent the experiences of other cultural backgrounds, linguistic backgrounds.

Third, this test was based on the Figma interactive prototype. So it cannot fully simulate the complete interaction, data updates and long-term user experience in a real application.

Fourth, the feedback for Kids Mode came from adult participants. Therefore, the actual appeal and comprehensibility of the Kids Mode still need to be further verified in subsequent ethically compliant studies.

Finally, the observer effect may exist during the testing process. Participants' awareness that they are participating in a test and being recorded may influence their natural usage behavior.

3 Results

3.1 Quantitative Results

Because this assessment primarily uses qualitative methods, so quantitative data is used more as supplementary reference.

3.1.1 Task completion status

Task	Completion Status	Description
Task 1: Browse community activities	6/6	All participants can browse the homepage and identify activities that interest them.
Task 2: View Activity Details	6/6	All participants can access the details page and understand the basic information.
Task 3: Experience scenario simulation	6/6	All participants completed the scenario simulation process and provided feedback on the feature.
Task 4: Check the Friends Participation	6/6	All participants understand that this page is related to the information of the activity participants.
Task 5: Experience Kids Mode	6/6	All participants viewed the children's mode and provided feedback from the perspective of a parent or family member.

Overall, the main task path of the Roota prototype is achievable. Participants were able to complete the 5 main tasks according to the test flow. This indicate that the basic interaction path of the prototype is understandable. But there are still some problems when they are on the interface. During the task completion process, participants raised several suggestions for improvement regard information completeness, button visibility, functional depth and child mode compatibility side.

3.1.2 Overview of High-Frequency Issues

Test records and observations show that some questions were repeatedly mentioned by multiple participants. These questions can be used for subsequent Problem Analysis.

Question	Situation	Explanation
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Details are not specific enough	many participants mentioned	Participants requested additional details on timelines, addresses, parking, items to bring and activities for children.
Simulation steps are few	many participants mentioned	Participants found the simulation features valuable but hoped for more steps, branches and dynamic feedback.
Kids Mode requires age-appropriate settings	many participants mentioned	Participants found the children's mode appealing. But different presentation methods might be needed for children of different ages.
Child lock	many participants mentioned	The current child mode is easy to accidentally triggering the parent mode.
Homepage activity discovery can be optimized	many participants mentioned	Issues included an insufficiently prominent "View All" button, a limited number of activities and a lack of personalized recommendations.

These results shows that Roota's core functional paths are generally understandable and achievable by users. But they still require more explicit and complete information support before making decisions. Especially the activity details page and scenario simulation function. Although the prototype receives positive feedback. But it still needs further improvement in information and interaction depth.

3.1.3 Quantitative data limitations

Due to the small sample size, the quantitative results in this study are only for supplementary explanation and are not suitable for statistical inference.

3.2 Qualitative Results

This section summarizes the themes of participant feedback based on the Think Aloud, semi-structured interviews and behavioral observation. After organize the test records, this assessment identified 6 main themes.

3.2.1 Theme 1: Activity information is generally understandable but lacks practical detail

Participants can understand the basic information on the activity details page. Such as activity type, time, location, suitable age group and activity nature. Color-coded labels also help users quickly determine whether the activity is suitable for their family.

“Based on this page, first of all, I can understand that it is a picnic and barbecue activity.” (P2)

However, participants repeatedly pointed out that while the current event details page was clear, it lacked specificity. Parents needed more practical details before going. Like check-in time, full address, parking information, navigation, items to bring and the specific content of the children's activities. P1 suggested adding both arrival details and children's activity details.

"It said 'Arrival and Check-in' here, but I think it would be better to also include the check-in time." (P1)

"For the 'Children's playtime,' I think you could add more details." (P1)

A similar issue appeared in P2 and P3's feedback. P2 feel that the activity process was too brief and needed a clearer timeline. And P3 emphasized some practical issues such as parking, navigation and items to bring. This suggests that the activity detail page should not only explain what the activity is, but also need help parents judge whether the activity it is convenient and realistic for their family to attend.

Therefore, the current activity detail page is effective in communicating basic information, but it does not fully meet parents' decision-making needs. For newly immigrated families, more specific event information can reduce uncertainty before participation and make the activity feel easier to join.

3.2.2 Theme 2: The homepage is visually attractive, but activity discovery can be improved

Most participants had a positive impression when they saw the homepage at first time. They felt the layout was relatively clear and the activity images is show a family-friendly, outdoor and community-oriented atmosphere in Sunday Picnic & BBQ.

"This page looks fairly clean and straightforward." (P1)

P1 also mentioned that the barbecue picnic activity on the homepage evokes images of family, children and weekend gatherings. This makes it quite appealing.

"It looks like a pleasant weekend activity." (P1)

However, the homepage still has issues with the efficiency of activity discovery. First, some participants felt that the number of activities currently displayed was too small and the categories were not rich enough. This might make the users feel that the information is limited. P2 also pointed out that the categories of nearby activities were not clear. So the users might not be able to quickly understand the logic behind when these activities were displayed. Second, the View All button is not eye-catching enough. P1 said she did not

notice the button during the test.

"I actually didn't notice this 'View All' option before. It's a bit too small." (P1)

Furthermore, the homepage recommendations need to be more personalized. P3 said that, his family are more prefer outdoor sports activities, so BBQ picnic may not catch his interest.

"Based on my personal preference, I think this football activity may be more interesting to me." (P3)

Therefore, while the homepage is effective in terms of visual appeal and basic information presentation. But the activity discovery process still has room for improvement. For future designs, can use method like increase the number of activities, optimize the category structure, improve the visibility of the "View All" option and provide more personalized recommendations based on family interests to improve user efficiency.

3.2.3 Theme 3: Scenario simulation reduces unfamiliarity but lacks interaction depth

The scenario simulation feature was one of the most positively received features in this test. Participants generally felt that it helped them understand the social situations that might occur during community activity in advance. And even reduced their uncertainty when participating in an activity for the first time. P2 states that her concerns about the activity mainly from a lack of understanding of the activity process. And scenario simulation can help her build psychological expectations in advance

"It can greatly reduce my fear of this activity." (P2)

"The fear mainly comes from my lack of understanding of this activity." (P2)

At the same time, the graphic and textual form of scene simulation and the selection mechanism also enhance the fun. P1 describes it as an experience similar to comics and mini games. This shows that the function not only has the function of information interpretation, but also increase users' willingness to explore activities.

"It gives a sense of pleasure, like reading a comic." (P1)

"The simulation... feels like playing a mini-game." (P1)

However, participants also believe that the depth of interaction simulated in the current scenario is still insufficient. The main problems include fewer steps, shorter interaction time, and insufficient impact of user choices on subsequent results. P2 believes that if different choices can affect the development of the subsequent plot, the simulation experience will be more meaningful

“The options in front of me will not affect the topic that appears behind me.” (P2)

“I think it might be better if it can affect him.” (P2)

In addition, P3 proposes add dynamic feedback such as character movement, button feedback or animation effects to enhance the immersion and interaction for user.

Overall, scenario simulation has validated its value in reduce unfamiliarity and establish activity expectations. But in the future, more branches, richer steps and dynamic feedback are still needed to add on.

3.2.4 Theme 4: Friends Participate helps build social security, but its influence varies between users

The Friends Participant feature can help users generate a sense of familiarity and social security. For some new immigrant families, participate in activities with acquaintances or same families can reduce psychological concerns during participation. Especially when users are not familiar enough with language, social scenes or activity processes. P2 has more significant positive feedback on this function. She believes that the involvement of acquaintances can lower her vigilance.

“Because of this acquaintance, I think my vigilance will also drop a little.”

(P2)

However, the impact of this feature on participation in decision-making is not consistent. Some participants believe that the peer list has some value, but it is more of auxiliary information rather than the core factor determining whether to participate in the activity. For them, whether the activity itself is attractive, is suitable to participate for their child and the overall activity experience are still more important. P1 describes this feature as adding icing on the cake.

“This feature is more of a nice-to-have addition.” (P1)

In addition, participants also acknowledged the preference tags in Friends Participant. For example, in BBQ activities, dietary preference information such as halal and vegetarian can help families avoid embarrassment or prepare inappropriate food in advance. However, such labels need to be adjusted according to different types of activities.

Therefore, the main value of Friends Participation lies in provide a sense of familiarity before the activity. But its impact on participation in decision-making will vary depend on user personality, language ability, acquaintance relationships and activity types. Subsequent designs should retain this feature while strengthening avatars, activity related tags, and privacy instructions to make it easier for users to understand.

3.2.5 Theme 5: Kids Mode has strong visual appeal but requires age-appropriate design and parental control

Kids Mode has received a lot of positive feedback. Participants believe that the colors, comic sense and picture book style expression in the children's mode are more suitable for children and can help them understand the activity content in a more relaxed way.

"It's the kind of style that children would love because it resembles comic strips or picture books." (P1)

On the other hand, participants also pointed out that Kids Mode needs to consider the preference differences of children in different age groups. Children in younger age groups may be more receptive to current picture book style content, but older children may require more mature or interactive ways of expression. P3 further suggests that static picture books may not be suitable for all children, and adding audiobooks, animations or animal characters may be more attractive for them.

"An audiobook might be a better option." (P3)

In addition, the child mode also needs to incorporate parental controls, especially child locks or mode locks. P3 believes that children are usually familiar with electronic devices and basic operations are not a problem, but if the child mode and adult mode are switched through only one button, children may accidentally touch and enter the adult interface. Therefore, participants suggested adding a mechanism similar to child lock to keep children in the version that suits them.

"Is it possible that you add a lock to this jump?" (P3)

Regarding the badge rewards, participants generally have a positive attitude, but also show caution. Badges can serve as positive incentives to help children gain a sense of achievement, but they should not be overused or addicted for the sake of rewards. P1 emphasizes that reward mechanisms should serve learning and participation, rather than simply chasing badges.

"The goal shouldn't be to keep watching just to earn badges." (P1)

Finally, the visual direction of Kids Mode is effective. But need further strengthen in three areas: adjust the content format according to different age groups, add expressions such as audio, animation or animal characters that are more suitable for children's attention characteristics and add child locks or parental controls to avoid children accidentally entering adult mode. Meanwhile, badge rewards should be used as moderate positive feedback rather than high-frequency, strongly stimulate gamified mechanisms.

3.2.6 Theme 6: Roota has practical value as a community activity aggregation platform

In summary, participants acknowledge the practical value of Roota as a community activity aggregation platform. For new immigrant families, community activity information is often scattered across different channels, such as schools, friend recommendations, community websites or offline announcements. Roota can integrate this information into one platform and provide activity details, scenario simulations and child modes. It will help reduce information search costs and make it easier for families to participate in local community life.

As P2 mentioned, currently the channels for families to obtain community activity information are relatively single, so an integrated platform would be of practical help.

“Our way of obtaining this information is still relatively single. Either through friends or school.” (P2)

P2 also believes that if Roota can combine activity information, scene simulation and children's picture book. It will help families better integrate into the community.

“If you could turn it into an integrated platform... I think that would be a really great idea.” (P2)

P3 also expressed a similar view. He thinks that Roota is particularly valuable for families who do not have time to search for activity information one by one. For busy parents, being able to browse, understand and judge whether activities are suitable for participation on one platform can reduce the additional burden of information search.

However, participants also pointed out that if they want users to use Roota for a long time, the product needs to be more refined and provide motivation for continued use. For example, points, badges or redemption mechanisms may increase users' willingness to continue participating in community activities, but these mechanisms need to be kept moderate to avoid making the product purely reward driven.

Therefore, the overall concept of Roota has gained initial recognition from users. It is not only a tool for displaying activity information, but also a supportive platform to help new immigrant families understand the activity process, build confidence in participation and enhance community connections. Future design should continue to strengthen information aggregation capabilities, activity screening efficiency and long-term participation motivation.

3.2.7 Theme 7: Interface labels and action wording caused uncertainty

Although most participants were able to complete the main tasks, some

interface labels caused uncertainty during the interaction. This issue appeared across different parts of the prototype, including the scenario simulation, Kids Mode, and check-in function.

For example, P4 was not sure what “Practice here” meant before entering the scenario simulation.

“I wasn’t very sure when I first saw this ‘Practice here’. I don’t understand what this means.” (P4)

A similar issue appeared in Kids Mode. P4 felt that the button label “Let’s go” was ambiguous because it was unclear whether the button meant starting the story, joining the activity, or making a participation decision.

“I just don’t know what that means. Does it mean directly participate? Or something else?” (P4)

The label “Check in” also created confusion. P4 initially understood it as a sign-up or registration button, rather than an action that should happen after the user arrives at the activity.

These examples suggest that button wording should be more specific. Since Roota is designed for newly immigrated families, unclear labels may increase uncertainty, especially for users who are already unfamiliar with local community activity processes.

4 Problem Analysis

4.1 Limited Homepage Discoverability

Some participants felt that the number of activities displayed on the homepage was limited, and some participants did not notice the "View All" entrance. If the homepage does not clearly guide users to view more activities, users may stop browsing or mistakenly think that there are not many activities provided by Roota.

4.2 Insufficient Activity Detail

The activity details page helps participants understand the basic information of the activity. They can see the type, time, location, suitable age range and the general flow of the activity. But participants still want to know more information, such as whether you need to register in advance, what to bring, whether there is a parking space, and whether there is public transportation nearby. If these problems are not explained, even if users understand the content of the activity, they still feel that they have not completed the preparation before participating.

4.3 Scenario Simulation Lacks Depth

Scenario simulation helps participants imagine the activity before going. It also made the activity feel less unfamiliar. The current scenario simulation process is short, and the selection will not affect what will happen next. Some participants expect these choices to bring different results or provide more detailed feedback. If the simulation

only shows a few basic steps, users still have questions about what to do in the actual situation.

4.4 Companion List Needs Stronger Visual and Privacy Design

The Friends Participate feature helped some users feel safer and less stressed about joining an activity, but the feature raised privacy questions. Some participants wonder if everyone could see their participation status or not. If this is not clearly explained, they might be concerned about using this feature.

4.5 Kids Mode Needs Age-Based Adaptation

Kids Mode made the activity easier for children to understand than a text-heavy activity page. Children could use this mode to explore which activities they were interested in.

Children of different age groups vary in terms of reading ability, comprehension skills, and interests. Younger children need simple stories, audio, pictures, or animation. Older children want more interaction. If Kids Mode uses one style for all children, it will only be valid for users of a specific age group. Parent control is also an issue in this mode. Children can view activities and show interest, but the final decision on whether to participate still should be made by the parents. One useful way is to let children choose activities they like and then send this interest to their parents. Some parts also need clearer wording. For example, a button such as “Let’s go” could mean starting the story or joining the activity. This kind of unclear label can cause confusion.

4.6 Reward System Needs Balance

Reward system needs balance because if users focus too much on earning points or badges, they may pay less attention to the activity itself. This is especially important in Kids Mode. Children enjoy rewards, but the goal should still be to help them understand the activity and feel ready to join. The leaderboard may also create pressure for some users because they might feel falling behind when they see other people have joined many activities.

5 Recommendations

5.1 Improve Homepage Activity Discovery

The homepage already provides several ways to find activities, including search, category filters, “For you”, “See all”, and “More nearby”. The issue is not that these tools are missing. The problem is that some discovery paths may still be easy to miss. The “See all” link should be made more visible. In the current design, it is small and placed on the right side of the section title. Some users may focus on the large activity card and not notice that they can view more activities. A clearer button such as “View all activities” or “See all nearby activities” could make this action easier to find.

5.2 Add Detailed Activity Timeline and Preparation Information

The timeline can be expanded with more practical details. Each step in the timeline could include a more specific time, location, and action. For example, for the “Sunday BBQ Picnic”, “Arrive and settle in” could show when users should arrive, where they should check in, and who they can ask for help. “Kids play area” could explain what

children will do, whether staff will supervise them, and whether parents need to stay nearby.

The page should also include a short preparation section before the activity. This section can tell users whether they need to register, what they should bring, whether food is provided, and whether the activity will continue if the weather changes.

Transport information should also be added, such as full address, parking, and nearby public transport.

5.3 Expand Scenario Simulation

At the moment, the choices seem to lead users through a similar path. The design could make different choices lead to slightly different responses or outcomes. This would make the simulation feel more meaningful and closer to a real situation.

The simulation could also include more common social moments. For example, it could show how to greet another family, how to join a group activity, what to do if the child feels shy, or how to ask the organizer for help. The entry label should also be clearer. Some users may not understand what “Practice here” means. A label such as “Preview the activity” or “Try the activity simulation” would better explain the function before users enter it.

5.4 Redesign Companion List

The companion list already shows useful social information, such as people the user knows, other attending families, family details, language tags, and food preference tags.

The main improvement should focus on privacy. Some participants wanted to know whether their own participation status would be shown to others. The page should include a short privacy note, such as “Only your friends can see your Going status” or “You can choose who can see your activity status”. This would make the feature feel safer. Users should also be able to control their visibility when they choose to attend an activity. For example, Roota could offer simple options such as “Show to friends”, “Show to attending families”, or “Keep private”. This gives users more control without making the process too complex.

The page could also explain why some families appear in the list. For example, “People you know” is clear, but “Also attending” could include a short note such as “Families who chose to share their status”. This would help users understand that the information is not shown without permission.

5.5 Improve Kids Mode with Age-Based Design

To make Kids Mode suitable for different age groups. Younger children may need more pictures, audio, and simple instructions. Older children may prefer more interactive content and less childish visuals. Roota could offer age-based versions, such as “Ages 3–6”, “Ages 7–9”, and “Ages 10–12”, so the content matches the child’s reading level and attention span. Children can use Kids Mode to explore the activity and show what they are interested in, but parents should still make the final decision. A button such as “Tell my parent” or “I’m interested” could let children share their interest with parents without directly joining the activity.

Parental control should be clearer. Roota could add a child lock or a simple parent confirmation step before children leave Kids Mode or move to the adult interface.

This would reduce accidental switching and make the mode safer for children to use.

5.6 Balance Badge Rewards and Motivation

The reward system should be used as gentle encouragement, not as the main focus of the app. Points, badges, and rewards can help users feel that their participation is noticed. For example, badges could be given for attending a first activity, trying a new activity type, or completing an activity preview. This would make the rewards support the purpose of Roota instead of turning the app into a game.

For children, rewards should be simple and positive. Badges can help children feel proud, but they should not encourage children to keep using the app only to collect rewards. The reward system should help children understand activities and feel ready to join them.

6 Conclusion

This usability evaluation shows that Roota is useful for helping newly immigrated families find and understand local community activities. All participants completed the main tasks, so the basic flow of the prototype was clear.

Users liked the activity timeline, scenario simulation, Friends Participate feature, and Kids Mode. These features helped them understand what would happen, feel less unsure, and prepare before joining an activity.

The main issues were also clear. The homepage needs a more visible way to explore more activities. The activity detail page needs more practical information, such as registration, parking, transport, and what to bring. Scenario simulation needs more steps and more meaningful choices. The companion list needs clearer privacy settings. Kids Mode needs age-based content and stronger parent control. The reward system should motivate users without creating pressure.

Overall, Roota has strong potential as a support tool for newly immigrated families. Future design should focus on clearer information, better preparation support, safer privacy control, and a more suitable child experience. Future testing should also include more diverse families and, if possible, direct feedback from children.

7 Appendix

Appendix A: Semi-structured Interview Questions

The question framework is as follows and can be freely adjusted.

A. Overall Experience

Overall, how did you feel about using this app today? What impressed you the most?
What parts of the app make you confused or left you uncertain during your experience?

B. Scenario simulation function

After experiencing the scenario simulation, has your understanding of the event changed?
Does this feature make you more willing to participate in the activity? Why?

C. Friend participation feature

What was your first reaction after seeing the notification that your friend had participated in the activity?
Did this feature influence your decision to participate? Please describe it.

D. Kids Mode

How do you think your child will react when they see this interface?
Would you allow your child to use this mode to browse activities on their own?

E. Future usage intentions

If this app is officially launched, will you use it to learn about and participate in community activities?
What features do you think need improvement to make you more willing to use it?

Appendix B: Test Script

Opening welcome

Hello, welcome to today's user test! Thank you so much for taking the time to participate!

We are testing a prototype app designed to help new immigrant families understand and participate in community activities. We invited you here today to hear your honest feedback and help us improve the design.

A few points to clarify first: We are testing the design itself, not you—so there are no right or wrong answers, and any reactions or thoughts you have are extremely valuable to us.

During the test, please try to express your thoughts verbally, such as what you are looking at, what you are thinking, what you find convenient or confusing, as if you are talking to yourself.

With your consent, we will record today's test. The recording will only be used for research and analysis and will not be made public. You can choose to pause or exit at any time.

Okay, let's begin. Feel free to ask me any questions.

Task 1: Browse community activities

(Hand the device to the participant and open the App homepage)

Now, please browse this page as you would any other app and find a community activity that interests you.

Remember to share what you see and think.

Task 2: View Activity Details

Okay, now please click on the activity you just saw, take a look at its contents and then tell me:

What is this activity about? Who is it suitable for? What do you think will happen if you attend?

Task 3: Experience scenario simulation

Next, you'll see a Scenario Simulation feature. Please click to try it out.

Afterwards, tell me your impressions—has your understanding of this activity changed?

Task 4: Check the "Friends Participation" prompts

Now please look here; there's a notification indicating that a friend of yours has also participated in this event.

Please tell me how you feel after seeing this information? Will it affect your decision to participate in the event?

Task 5: Experience Kids Mode

Finally, we have a version specially designed for children. Please switch to it and take a look.

You can tell us what parts attract you the most, what parts you think your child would like, or if there's anything that's not quite right.

Conclusion

Okay, that concludes today's task. Thank you so much for your participation!

Next, I have a few questions for you. We'll have about 5 minutes. Would you mind continuing our conversation if it's convenient?

(Proceed to Appendix A: Semi-structured Interview)

Appendix C: Task Observation and Score Record Form

Participant Basic Information

Participant ID	
Test Date	
Test Duration	
Immigration Background (Nationality/Place of Origin)	
Children number	
Children's Age	

Task 1: Browse community activities	
Observation Project	Records / Ratings
Can she/he independently find activities of interest?	☐ Yes ☐ No ☐ Guidance required
Places of pause or confusion	
Key message from Think Aloud	
Other observations	
Participant satisfaction rating	☐ 1 Very dissatisfied ☐ 2 Dissatisfied ☐ 3 Neutral ☐ 4 Satisfied ☐ 5 Very satisfied

Task 2: View Activity Details	
Observation Project	Records / Ratings
Can she/he accurately describe the content of the activity?	☐ Yes ☐ No ☐ Part
Do she/he understand who this activity is suitable for?	☐ Yes ☐ No ☐ Part
Is there a misunderstanding of	☐ Yes, where/why

information?	☐ No
Key message from Think Aloud	
Other observations	
Participant satisfaction rating	☐ 1 Very dissatisfied ☐ 2 Dissatisfied ☐ 3 Neutral ☐ 4 Satisfied ☐ 5 Very satisfied

Task 3: Experience scenario simulation	
Observation Project	Records / Ratings
Does she/he express a sense of unfamiliarity/reduced anxiety?	☐ Yes ☐ No ☐ No significant changes
Did she/he express your expectations for the activity's flow?	☐ Yes ☐ No
Does this indicate an increased willingness to participate?	☐ Yes ☐ No ☐ No change
Key message from Think Aloud	
Other observations	
Participant satisfaction rating	☐ 1 Very dissatisfied ☐ 2 Dissatisfied ☐ 3 Neutral ☐ 4 Satisfied ☐ 5 Very satisfied

Task 4: Check the "Friends Participation" prompts	
Observation Project	Records / Ratings
Do she/he understand	☐ Yes

the meaning of this function?	☐ No
Impact on participation decisions	☐ Positive ☐ No effect ☐ Negative
Key message from Think Aloud	
Other observations	
Participant satisfaction rating	☐ 1 Very dissatisfied ☐ 2 Dissatisfied ☐ 3 Neutral ☐ 4 Satisfied ☐ 5 Very satisfied

Task 5: Experience Kids Mode	
Observation Project	Records / Ratings
Do she/he actively click to explore content?	☐ Yes ☐ No
Mention the most attention-grabbing elements	
Do she/he express your willingness to let your child use it?	☐ Yes ☐ No ☐ Unsure
Key message from Think Aloud	
Other observations	
Participant satisfaction rating	☐ 1 Very dissatisfied ☐ 2 Dissatisfied ☐ 3 Neutral ☐ 4 Satisfied ☐ 5 Very satisfied

Task	Completion status	Satisfaction rate
Task 1: Browse community activities	☐ Completed ☐ Incomplete	
Task 2: View Activity Details	☐ Completed ☐ Incomplete	
Task 3: Experience scenario simulation	☐ Completed ☐ Incomplete	
Task 4: Check the "Friends Participation" prompts	☐ Completed ☐ Incomplete	
Task 5: Experience Kids Mode	☐ Completed ☐ Incomplete	
Overall satisfaction	☐ Completed ☐ Incomplete	

Supplementary Interview Record

Appendix D: Coding and Thematic Analysis Table

Theme	Codes	Evidence Source	Related Finding
Activity information detail	timeline, address, parking, what to bring, children's activities	P1, P2, P3, P5	Activity details were clear but lacked practical decision-making information
Homepage discoverability	View All visibility, activity quantity, personalisation	P1, P3, P5	Homepage was clear but activity discovery needed improvement
Scenario simulation	uncertainty reduction, mini-game feeling, lack of branch	P1, P2, P3, P4, P5, P6	Simulation reduced unfamiliarity but needed more depth
Friends Participation	social safety, familiar people, dietary tags, avatars	P1, P2, P3, P4, P6	Had social value but varied in decision impact
Kids Mode	visual appeal, age adaptation, audio, animation, child lock, badge concern	P1, P2, P3, P4, P5, P6	Kids Mode was attractive but needed age adaptation and parent control
Interface labels and	unclear labels	P4	Some button labels

action wording			caused uncertainty because users were not always sure what would happen after clicking.
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